Rounak Dey

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Education

Indraprastha Institute of Information Technology (IIIT Delhi) — New Delhi, IN
Bachelor of Technology in Computer Science & Engineering — *Aug. 2023 – 2027*

Technical Skills

Languages: C/C++, Java, Golang, SQL
Competitive Programming: Leetcode Knight (1977), Codeforces Expert (1671)
Expertise: Data Structures & Algorithms, Operating Systems, System Design, Object Oriented Programming
Tools: Git, Docker, Linux, Kafka, Kubernetes, Redis, RabbitMQ, Elasticsearch
Databases: MongoDB, DynamoDB, Cassandra, PostgreSQL, MySQL
Coursework: Introduction to Fintech, Operating Systems, Database Management Systems, Computer Networks

Projects

Verdis-Go: Redis-Compatible In-Memory Data Store | *Golang, TCP/RESP Protocol* — GitHub | 2026

- Engineered a Redis-compatible in-memory data store in **Go** from scratch, implementing the **RESP protocol** and **25+ commands** across strings, lists, hashes, and sets, sustaining **79K req/s (GET)** and **77K req/s (SET)** at sub-millisecond p99 latency
- Built a **goroutine-per-client** concurrency model with **16-way sharded, fine-grained locking**, sustaining **60-71K req/s at 500 concurrent clients** with consistent throughput across all data types
- Implemented dual persistence (**AOF** command logging + **RDB** snapshotting) at a measured **14% throughput cost**, and **master-replica replication** with full/partial resync via a replication backlog, verified through live crash-recovery testing

CipherChat | *C++17, Socket Programming, OpenSSL* — GitHub | 2025

- Built a real-time **encrypted** client-server chat platform in **C++17** supporting secure multi-room communication over **TCP**
- Implemented **X25519** key exchange, **HKDF-SHA256** key derivation, and **AES-256-GCM** authenticated encryption with AAD-bound per-room and per-session keys
- Developed a **multithreaded** server architecture (accept thread + worker-per-client) with dynamic room management, encrypted message history replay, and a custom binary framing protocol

Pesky Birds | *Java, LibGDX, Box2D, Gradle* — GitHub | 2025 – 2026

- Developed a **2D physics-based** Angry Birds clone using **LibGDX** and **Box2D**, featuring slingshot-driven projectile mechanics and collision-based destructible environments
- Built multiple levels with progression, scoring, win/lose conditions, pause/resume, and a level-selector system
- Designed a modular **OOP architecture** applying inheritance (shared `GameObject` base class), polymorphism (overridden `update` methods), abstraction (`Renderable` interface), and encapsulation across game entities

Experience

Undergraduate Student Researcher — August 2025 – Present
Network Security Lab, IIIT Delhi — *New Delhi, India*

- Conducted dynamic behavioral analysis of Android malware using the **COMEX** testbed across **11 real Android devices**, collecting **network**, **syscall**, and **CPU** runtime data
- Built a **TF-IDF** syscall feature pipeline (**106K+ features**, **19.5K samples**) and trained an **MLP classifier** achieving **82.7% accuracy** for malware detection
- Reproduced **Hfinger**-based HTTP classification on **406K+ requests**, showing syscall features outperform network-based features under **open-set conditions**

Achievements

LeetCode: Solved **300+** problems, achieving a highest rating of **1977 (Knight)**.
Codeforces: Achieved a highest rating of **1671 (Expert)**, Ranked 1550 in Codeforces Round 1117 (Div. 2)
JEE Main 2023: Secured a position in the top **1%** of over 1 million candidates nationwide.
JEE Advanced 2023: Achieved All India Rank (AIR) **11,800**, placing among the top-performing engineering aspirants in India.